RETAIL SALES DATA ANALYSIS REPORT

INTRODUCTION

This report analyzes sales performance, customer demographics, and purchasing behavior using historical sales data. The objective is to identify trends, key customer segments, and product performance to support data-driven decision-making.

1. Data Loading and Cleaning

The dataset was load on and cleaned with Microsoft Excel Application. The file was initially on Comma Separated Value (CSV) format. The dataset was professionally cleaned for easy Exploratory Data Analysis (EDA) to generate insightful reports.

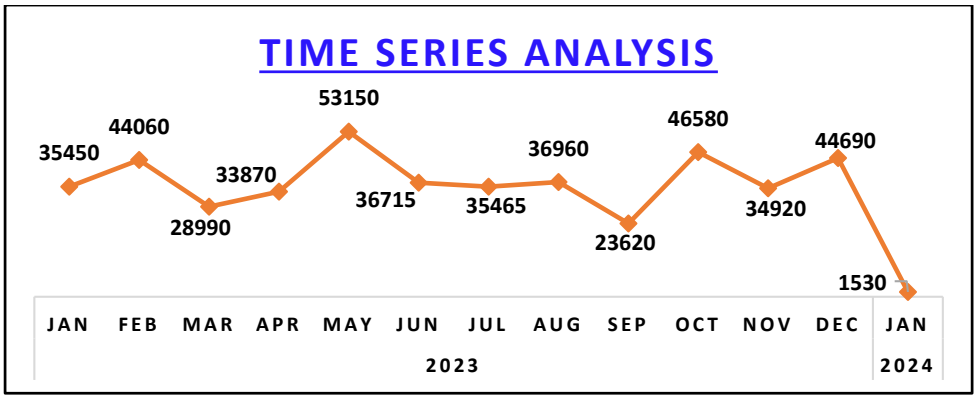
2. Descriptive Statistics

Basic statistical calculations were performed to determine the Mean, Median, Mode, and Standard Deviation of key detailed columns. Below is the analyzed chart representation.

METRIC	Age	Quantity	Price Per Unit	Total Amount
Mean	41	2,514	179,890	456,000
Median	42	3,000	50,000	135,000
Mode	37	4,000	50,000	50,000
Standard Deviation	14	1,132	189,586	559,718

3. Time Series Analysis

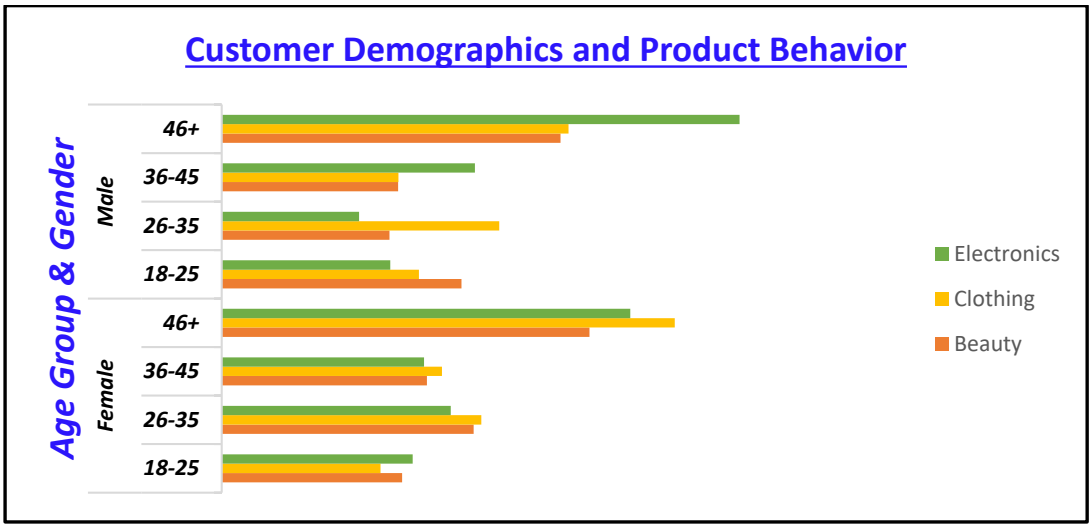
Sales data was analyzed over time using time series techniques such as a pivot table, line chart, and data labels. Dates were grouped by month to observe trends, and data labels were added for easy understanding. Sales fluctuate over the year, however, there was a peak in the month of May, 2023. There was a drastic dropped in January, 2024. Below is the visualized representation generated with a line chart.



4. Customer and Product Analysis

a. Customer Demographics

This analyzes customer's characteristic towards sales and how it affect the trends. It gives insight on which category patronizes most and which product catches their interest. From the visual below, it is visible that male gender shop more, people in age group 46+ (46 years and above) have money to spend, and the product Electronics generate most sales. A bar chart was used to generate the visual.



b. Purchasing Behavior

This analyzes uses heat maps (color scales) to show which product generate more income; the green signifies the maximum sale while the red color signifies the minimum sale. Electronics recorded the highest sales, indicating strong customer preference.

Additionally, the average spending analysis shows that age group 46+ spends more per transaction. Below is the heat map visual.

PRODUCT BEHAVIOR HEAT MAP				
Age Group/Gender	**Beauty**	**Clothing**	**Electronics**	**Grand Total**
18-25	**31280**	**26510**	**26760**	**84550**
Female	13435	11825	14210	39470
Male	17845	14685	12550	45080
26-35	**31240**	**39975**	**27265**	**98480**
Female	18750	19320	17045	55115
Male	12490	20655	10220	43365
36-45	**28405**	**29550**	**33915**	**91870**
Female	15270	16400	15065	46735
Male	13135	13150	18850	45135
46+	**52590**	**59545**	**68965**	**181100**
Female	27375	33730	30415	91520
Male	25215	25815	38550	89580

RECOMMENDATIONS

- Focus marketing efforts on 18 – 25 age group.
- Increase inventory and promotions for high-performing product (Electronics).
- Plan ahead for peak sales periods to maximize revenue opportunities.

CONCLUSION

The analysis highlights key drivers of sales performance, particularly within specific customer segments and product categories. By leveraging these insights, the business can optimize marketing strategies, improve customer engagement, and enhance overall profitability.